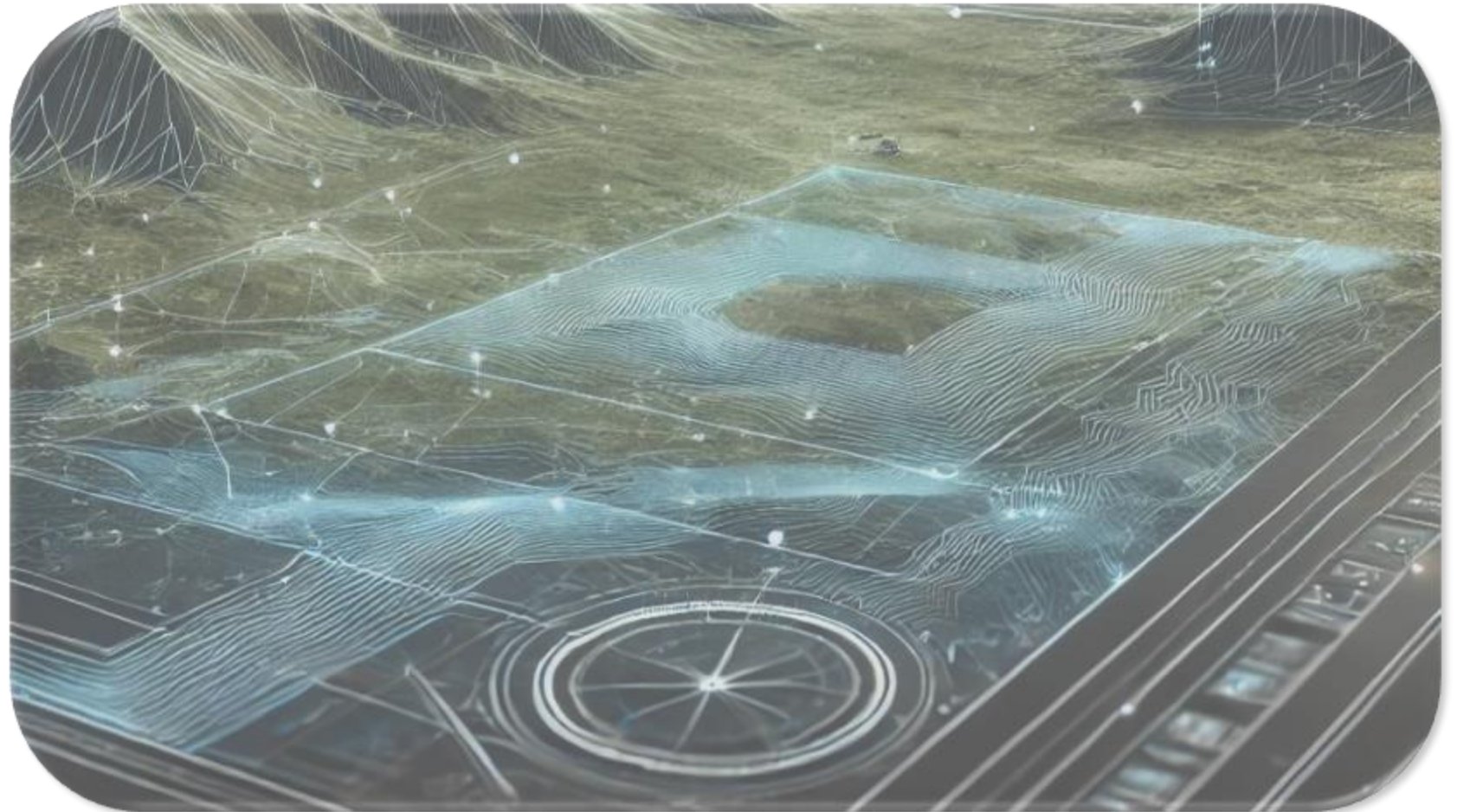


Proposed Enhancements for Military Low Overhead Models

Final Presentation

Team D

- Joshua Sourbeer
- Phani Raja Bharath
- Michael Cuddy
- Steven Thompson



Section 1: Defining the Problem & Simulation Background

- Review of Project Goals
- Relevant Research and How it Supports Us

Section 2: Recommended Solutions (Josh)

- 3D Terrain Incorporation
- Improved Pathfinding Ability

Section 3: Improved Game Logic (Bharath)

- Core Entities
- Finite State Machines

Section 4: Integrating Opposing Force AI (Mike)

- How it works
- Visualization in Action

Section 5: Large Language Model (Steven)

Problem #1: Inclusion of Modern QOL Smart Features

- Ability to quickly sequence commands (Move+Breach+Attack+Seize+Defend)
- Ability to Disaggregate and Reaggregate Units for realistic engagements
- Develop a finite-state machine to encompass necessary minor commands
 - Pathing between two points, machine accounts for the path chosen and obstacles encountered

Problem #2: Lack of Terrain Detail

- Ability to consider elevation in real time
- Ability to consider more than one route around pieces of terrain to arrive at a set destination (consider shortest distance, shortest time, fewest obstacles)
- Referencing the unit's ability to interact with the terrain

Problem #3: Integration of AI Assisted Opposing Force

- Determine a viable method to enable an agent based intelligent OPFOR entity that assists human World Class OPFOR operators.

Problem #4: Opposing Force Behavior

- Develop a Program that can reference Enemy Doctrine, and Note specific tactics associated with specific Nations

Riotto, Teaching the Army

- Supports our work by suggesting a 3D, Modular learning environment (VBS3) is preferred over the current Low Overhead Models (DXTRS)

Naylor, Evaluation of DXTRS for Sustainment

- Supports our work by showing the gaps DXTRS has in intelligent logistical play and features.

The Army's AI Strategy

- Supports our research by emphasizing that AI will be seen and viewed as a partner on the battlefield.

What are Some Common Features of Low Overhead Models?

- Simulations that typically train Individual Soldiers-Brigade Sized Elements (3,000-5000 Soldiers)
- Typically, can transmit simulation data to real world Military computerized Mission Command Systems
- Often consist of a Map based interface, Military unit control, and some system to manage indirect fires, combat operations, supply concerns and others
- Each has its own advantages and disadvantages; Hawaii uses JCATs extensively for exercises throughout the Pacific so partners can operate the system, but it is rarely used in CONUS because it is a very outdated system and not an official Program of Record for Army usage

Simulation	Simulation Type	Proponent	Target Audience	Notes	Accredited?
DXTRS	Constructive	CAC	All Up to Brigade	2D Environment	Yes, Army
LOTS	Constructive	JTDS	All Up to Brigade	Better Intel Play than DXTRS	Yes, Joint
VBS3	Virtual	Games for Training	Individual-Platoon	3D Environment, Limited Terrain	Yes, DOD Wide
JCATS (JLOD)	Constructive	JSJ7	Joint/Multinational		Yes, Joint

What are Some Common Limitations of Low Overhead Models?

- They lack Cloud Based features and all have hardware requirements which Soldiers must meet to use the system
- They cannot be used outside training events
- They lack the combination of 2D & 3D interface for full immersion while battle tracking
- Terrain Visualization is only expertly demonstrated in VBS3, which has limitations on how large the map can be, and how large of an exercise can be trained based on locally hosted servers.
- Only VBS3 has automated features for managing adjacent, non-critical units; and VBS3 plays at the entity level. So a company icon in DXTRS consists of 200 individual Avatars in VBS3.

Simulation	Main Limitations
DXTRS	Doesn't stimulate every MCIC, difficult to use without MTC assistance, Lengthy Mission build period
LOTS	Joint POR not Army, so there is less backside support and interative development. It is popular for specific events in specific areas.
VBS3	Not scaled to support larger exercises on large maps with thousands of entities
JCATS (JLOD)	It can't be found everywhere and not many CONUS organizations are trained on it

Verification, Validation, Accreditation: Major Stakeholders

Major Stakeholders During Army Simulation Development

Generating Requirements



NSC, COEs, FORSCOM, etc.

Funding the Process



Army G8

Building the Simulation



Industry Partners (Lockheed, Cole Engineering, Bohemia)

Verification & Validation



PEO-STRI Oversees Industry,
NSC Monitors PEO STRI's
V&V Milestones

Accreditation Authority



National Simulation Center

Fielding & Distribution



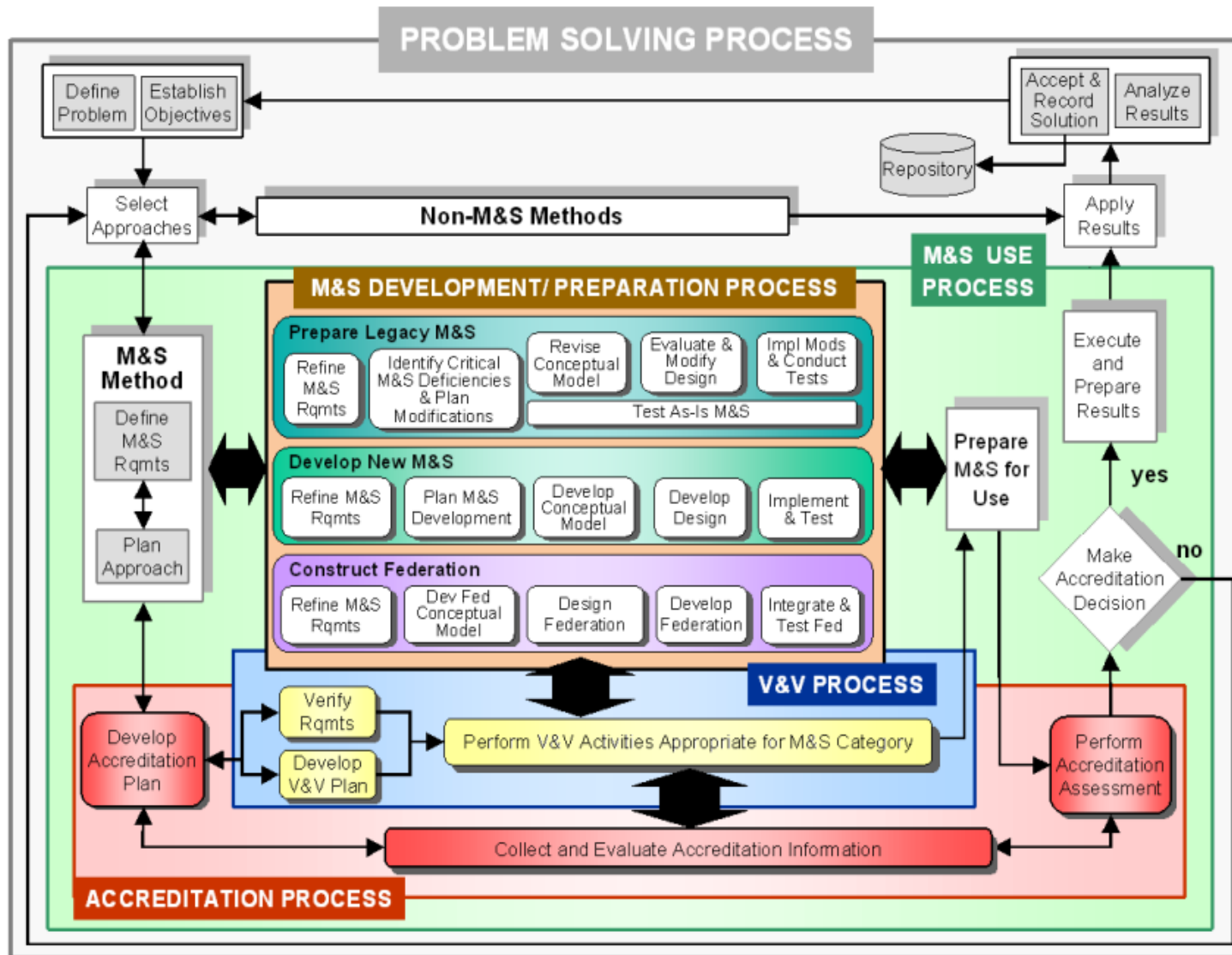
Training & Maintaining



Battle Labs, MTCs

Verification, Validation, Accreditation: Best Practices

Source: DOD VV&A Recommended Best Practices Guide

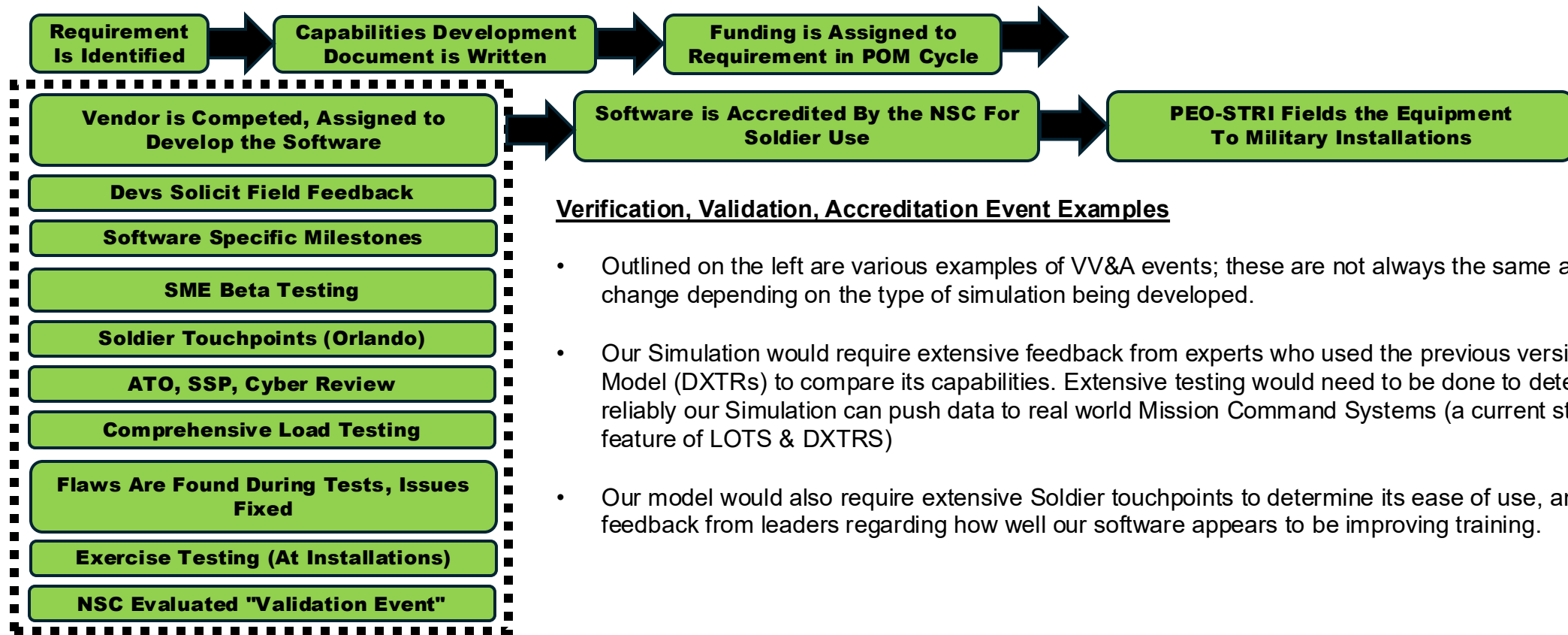


The Overall Problem Solving Process

Our Model's VV&A Process

- **Verification:** Does our Model Accurately Reflect the Original Specifications for the Design?
 - In this case our model would be tested extensively by SMEs during various Milestones in VV&A to Check if we had successfully met all the original requirements outlined in the Capabilities Development
- **Validation:** Does our Model Accurately Reflect the Real-World System?
 - In our case, the real-world system would be movement speeds, combat engagements, indirect and direct fires, enemy behaviors. So our model would be validated to see how accurately it depicts real world combat.
- **Accreditation:** Is our model officially certified and approved for DOD Use?
 - Our Model would be Accredited for DOD Use by the National Simulation Center (Fort Leavenworth, KS) as a final accreditation event with Soldiers. If our model fails accreditation, fielding will be delayed as more changes would be required.

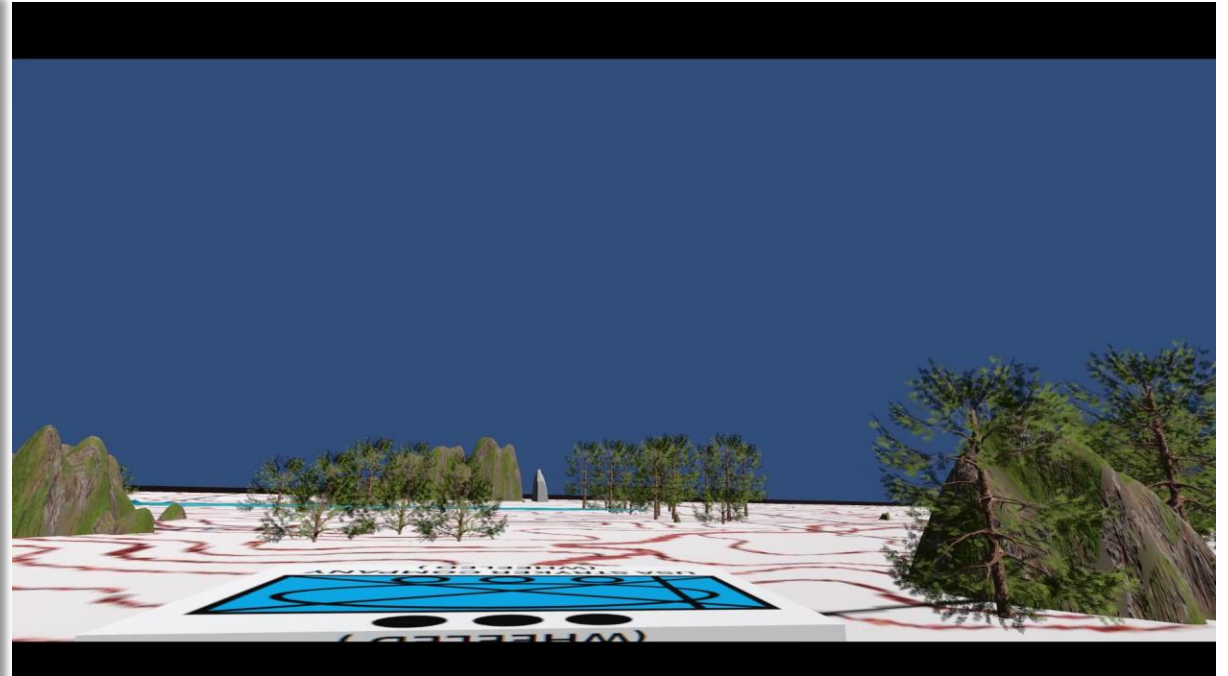
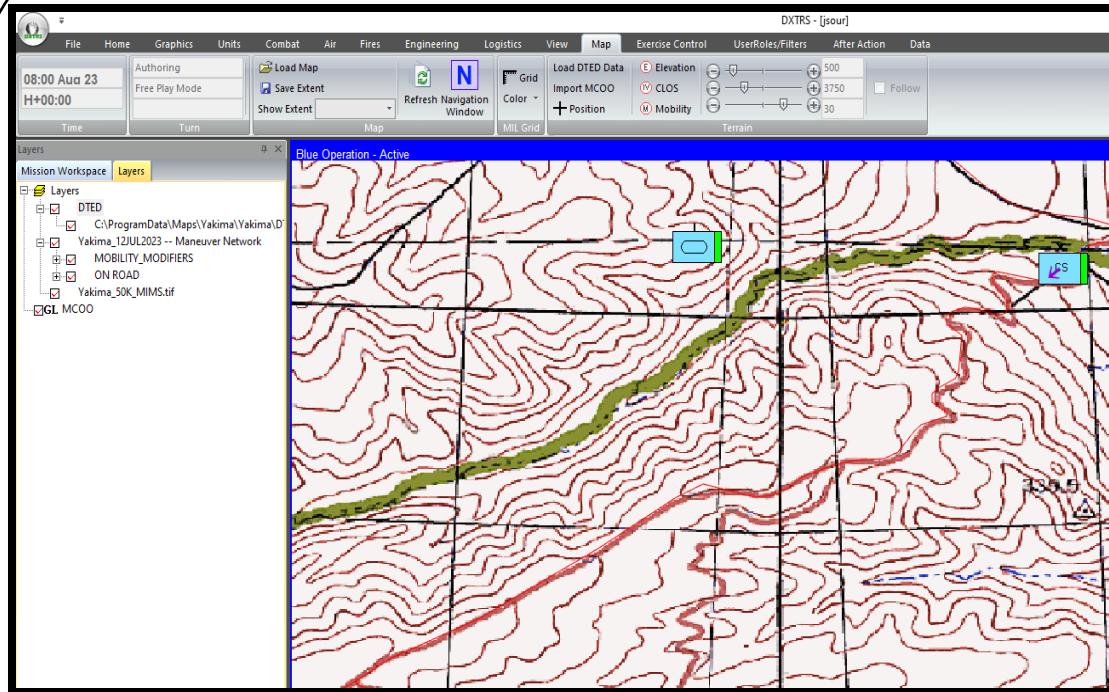
Verification, Validation, Accreditation: Examples of VV&A Milestone Events



Verification, Validation, Accreditation Event Examples

- Outlined on the left are various examples of VV&A events; these are not always the same and often change depending on the type of simulation being developed.
- Our Simulation would require extensive feedback from experts who used the previous version of our Model (DXTRs) to compare its capabilities. Extensive testing would need to be done to determine how reliably our Simulation can push data to real world Mission Command Systems (a current staple feature of LOTS & DXTRS)
- Our model would also require extensive Soldier touchpoints to determine its ease of use, and feedback from leaders regarding how well our software appears to be improving training.

Improvements to Terrain Visualization



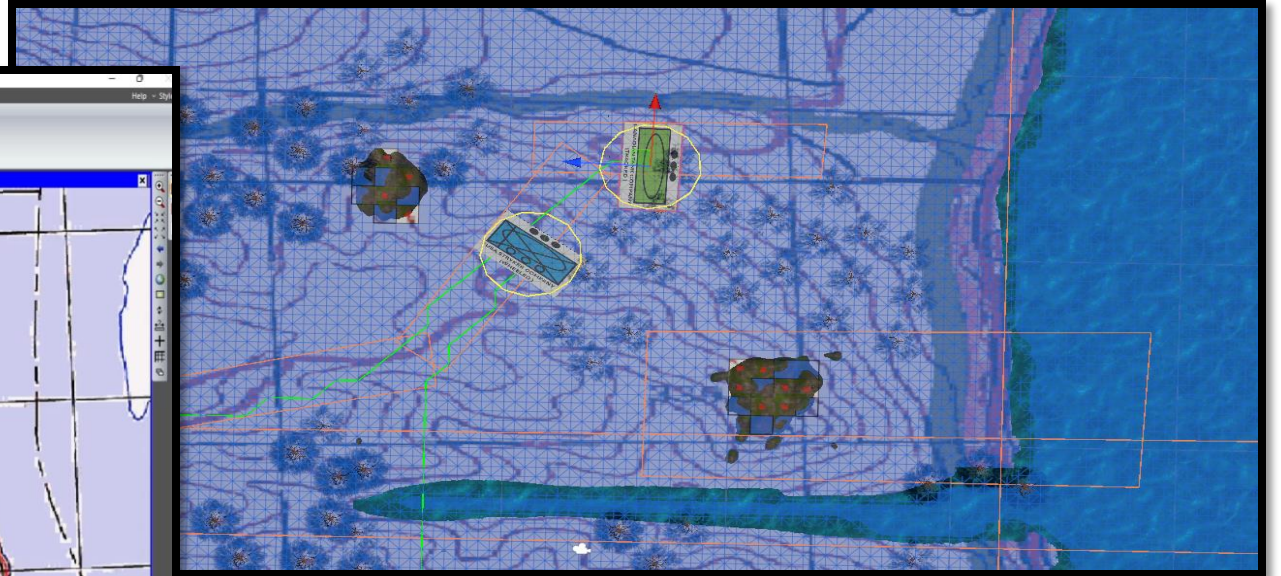
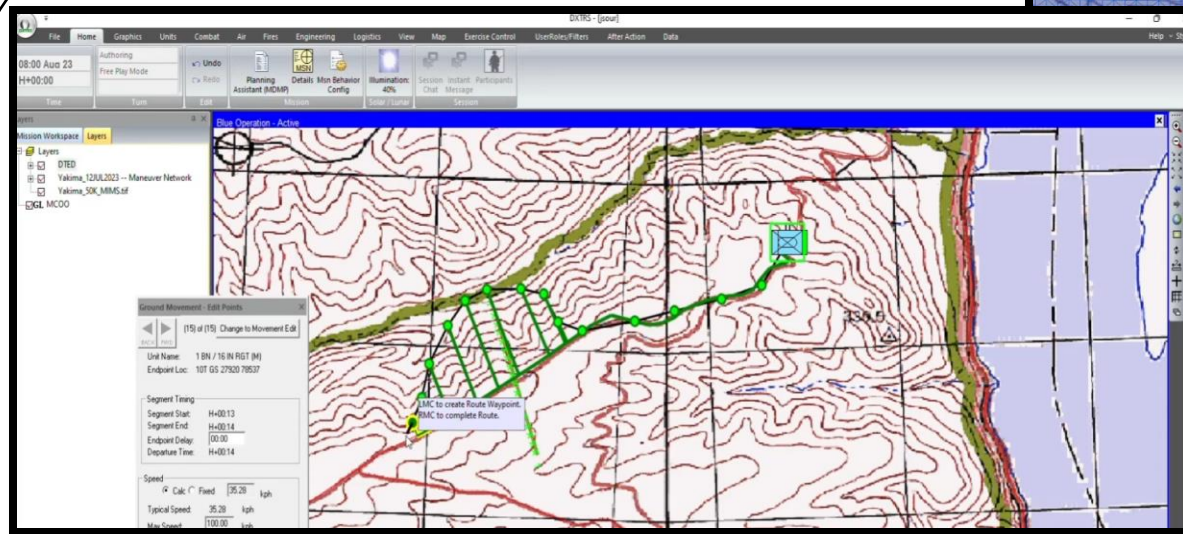
DXTRS Terrain Visualization

- 2D Map Based
- Requires Multiple Layers to be Externally Downloaded and Added for Terrain Data (DTED Format) to be Recognized by DXTRS
- Terrain Packages are not always provided at the time of fielding

Hybrid 2D-3D Terrain Visualization

- Incorporates Base MIM (Military Installation Map) with 3D Objects over Terrain for First Person Navigation.
- Only Requires the Base MIM, 3D Objects and Terrain Layer. These are all built into the Game and Don't Require External Loading.
- Allows for improved way to see obstacles in the path of movement

Improvements to Pathfinding



DXTRS Pathfinding Issues

- Automatic route preferencing for hardball roads is hardcoded; cannot be adjusted
- Obstacle auto-navigation is limited to minefields, existing bodies of water
- Units behave differently, but this is hard coded based on the task organization and cannot be manipulated
- Scripted movements for non-essential units take up valuable time

Our Model Improved Features

- Zones can be drawn on the map where navigation is more difficult-this can be done during the game
- Units can be given different movement characteristics
- Everything can be changed on the fly, during run time

DXTRS Pathfinding Comparison Matrix

DXTRS Pathfinding Ability Compared With Our Version

Goal of the Comparison Metric	Comparison Metric	DXTRS	Our Version
Flexibility During Run Time	Change the Destination While Sim is Running	X	Yes, Change Destination During Runtime
Flexibility During Run Time	On the Fly Map Changes	X	Yes, Add 3D Terrain to Change Grid
Capability Changes	Change Capabilities of Units	Yes, But Requires Code Manipulation	Completely Agnostic
Grid Comparison	Box Grid Nodes	1,147	1,394
Productivity of Searching	Searched Nodes	All 1,147	Only 64-466 At a Time Every 2 Seconds
Searching Type	Unknown	Passive	A*
Ease of Setup	Setup Procedures	Load DXTRS>Load DTED (Terrain)>Load Task Organization>Select Units>Add Units to Scenario>Drag Unit to Map>Right Click, Select Movement>Select Destination>Add to Timeline Repeat Steps 7-9 for Each Subsequent Movement	Load Unity Project>Attach Target Game Object to the Unit Icon>Move to Desired Destination as Required
Simulation Aesthetics	3D Terrain Visualization	X	Yes, Hybrid 2D-3D
Automated Obstacle Interactions	Automatic Bypass	Only Works for Minefields and Natural Water Obstacles	Works for Whatever We Designate as Impassable

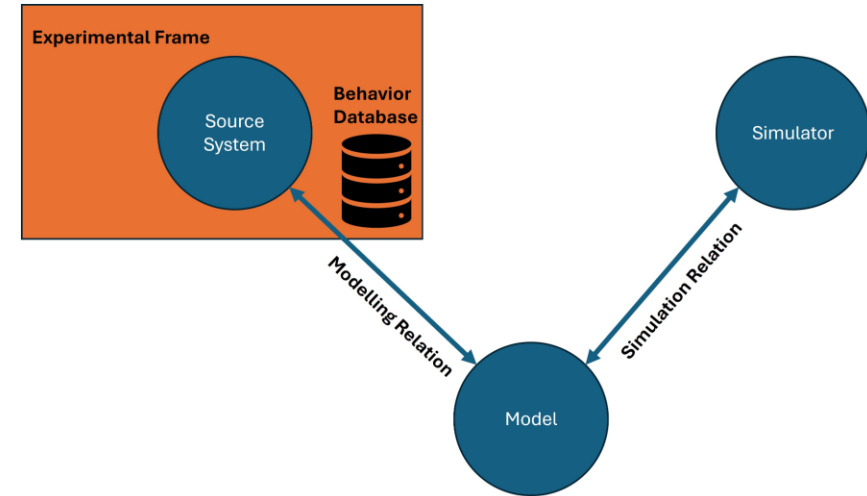
Core Entities of the MSF, Simulation Type, Method

Our Proposed Model

- A revised and improved low overhead model, capable of real time changes, improving the opposition force/partner force behavior, and a better visualization of 3D terrain on a 2D map.

Core Entities of the Modeling and Simulation Framework

- Source Systems:
 - Human Input: Database Build (Task Org)
 - Game Mechanics: Fires, Sustainment/Log, Combat
- Model: Ground Model & LLM
- Simulator: Computer Simulation



Simulation Type: Simulation by Experience Used for Training decision making

Method: Hybrid Agent Based

Simulation Element	Continuous	Discrete
Automated Movements/Activities	✓	
Timeline Events		✓
Enemy OPFOR Activities	✓	

Deep Reinforcement Opponent Network (DRON)

AI Concept Used

How it Works

Visualization in Action

Verification, Validation, Accuracy



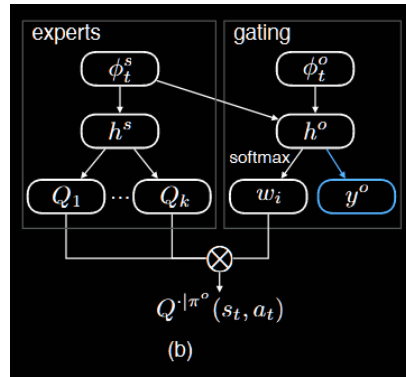
Deep Reinforcement Opponent Network (DRON) Model

Agent Based

Deep Q Neural Network

DRON considers the current state of the Agent, the actions of multiple Agents around it, and predicts the best action to take before transitioning to the next state.

How it Works



INPUT LAYER

EXPERTS

Sensing Data

Agent types	Agent State
Unit Designation	Stationary
Unit Type	Moving
Friendly or Enemy	
Relation to Self	Agent Actions
Adjacent Unit	Engaging
Higher HQ	Defending
Subordinate Unit	Retreating
	Establishing
	Movement to Contact

Information

GATING

Observations

- Past Unit States
- Past Unit Actions

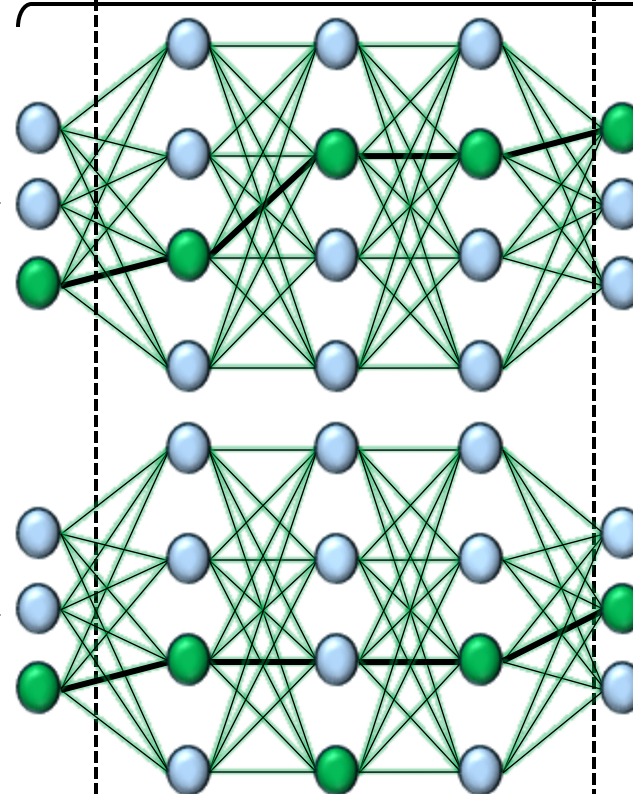
Information

HIDDEN LAYER

OUTPUT LAYER

$$Q^{\pi}(s, a) \equiv \mathbb{E} [\sum_t \gamma^t r_t | s_0 = s, a_0 = a, \pi]$$

DRON (π POLICY)



Q-Values

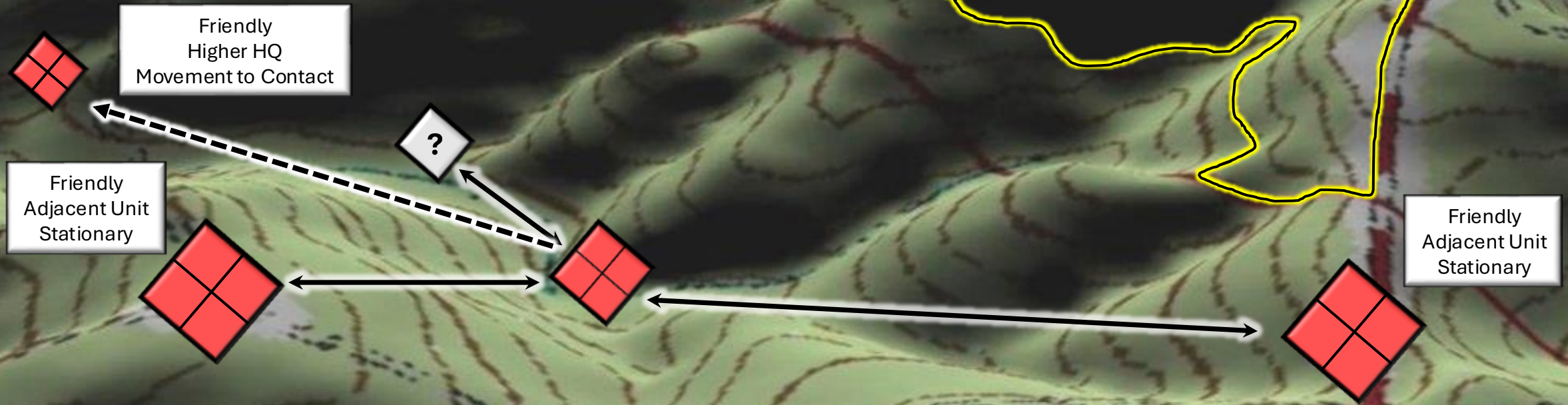
Q-TABLE

	A	A	A
S	#	#	#
S	#	#	#
S	#	#	#

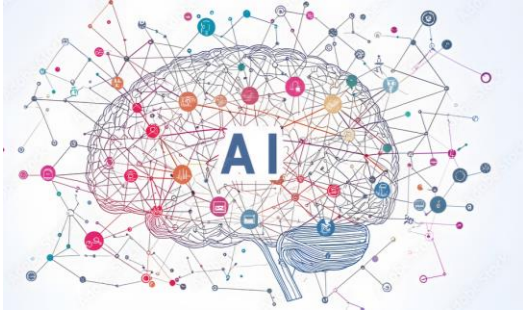
IDEAL ACTION

Visualization

Our agent then determines a route based off A* to move to the proximate zone of support



DRON (π *POLICY*)



Accuracy? $\updownarrow$

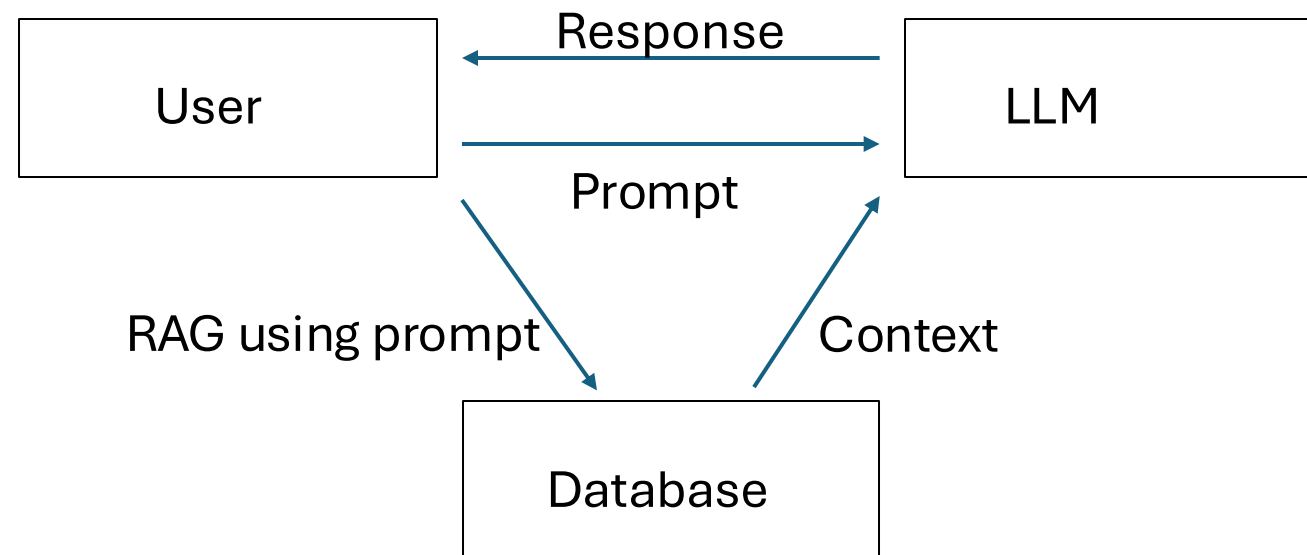
DRON (*TARGET POLICY*)



Model	Data	Software and Implementation	Human
• Simplification • Omission • Incorrect Relationships	• Input Data • Data Interpretation	• Coding • Numerical Errors	• Cognitive Biases • Communication Errors

- Does the LLM provide the correct relationships between OPFOR units as input data?
- Are we pulling the correct input layer data from the simulation real time in reference to units, types, actions, and states? Is it in a readable format that represents the data accurately?
- Is the algorithm producing the correct predictions of OPFOR movement based on Human World Class OPFOR actions?
DRON π *POLICY* vs. DRON *TARGET POLICY*
- Is our Target Policy informed by experts without cognitive bias?

- Retrieval Augmented Generation (RAG)
 - RAG acts as a framework that LLM to generate responses with
 - Based on prompt given, searches the documentation of relevant text
 - Text returned to LLM to provide the AI context and information
 - Minimizes hallucination and maximizes answer relevance



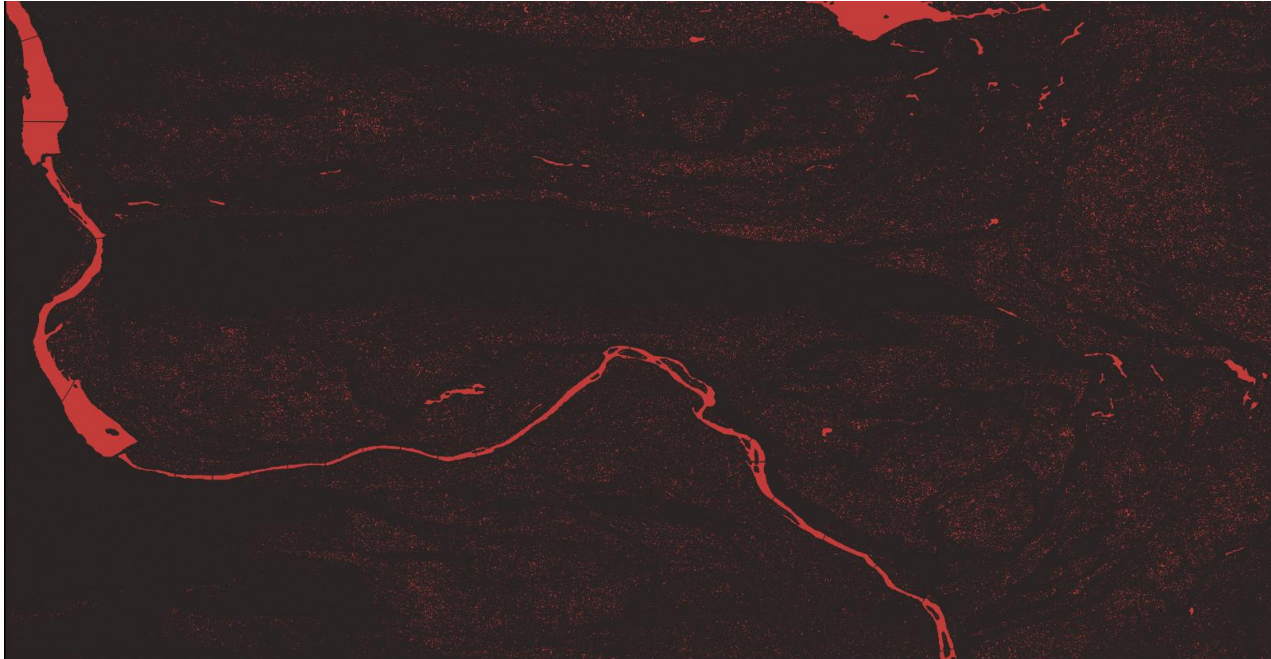
- LLM utilized: GPT 3.5 Turbo
 - Chosen for being fast and low cost
- Database service: Datastax
- LLM modification: Retrieval Augmented Generation using Langchain

Response	LLM with RAG	LLM without RAG	Random prompts
Score	0.941	0.921	0.873

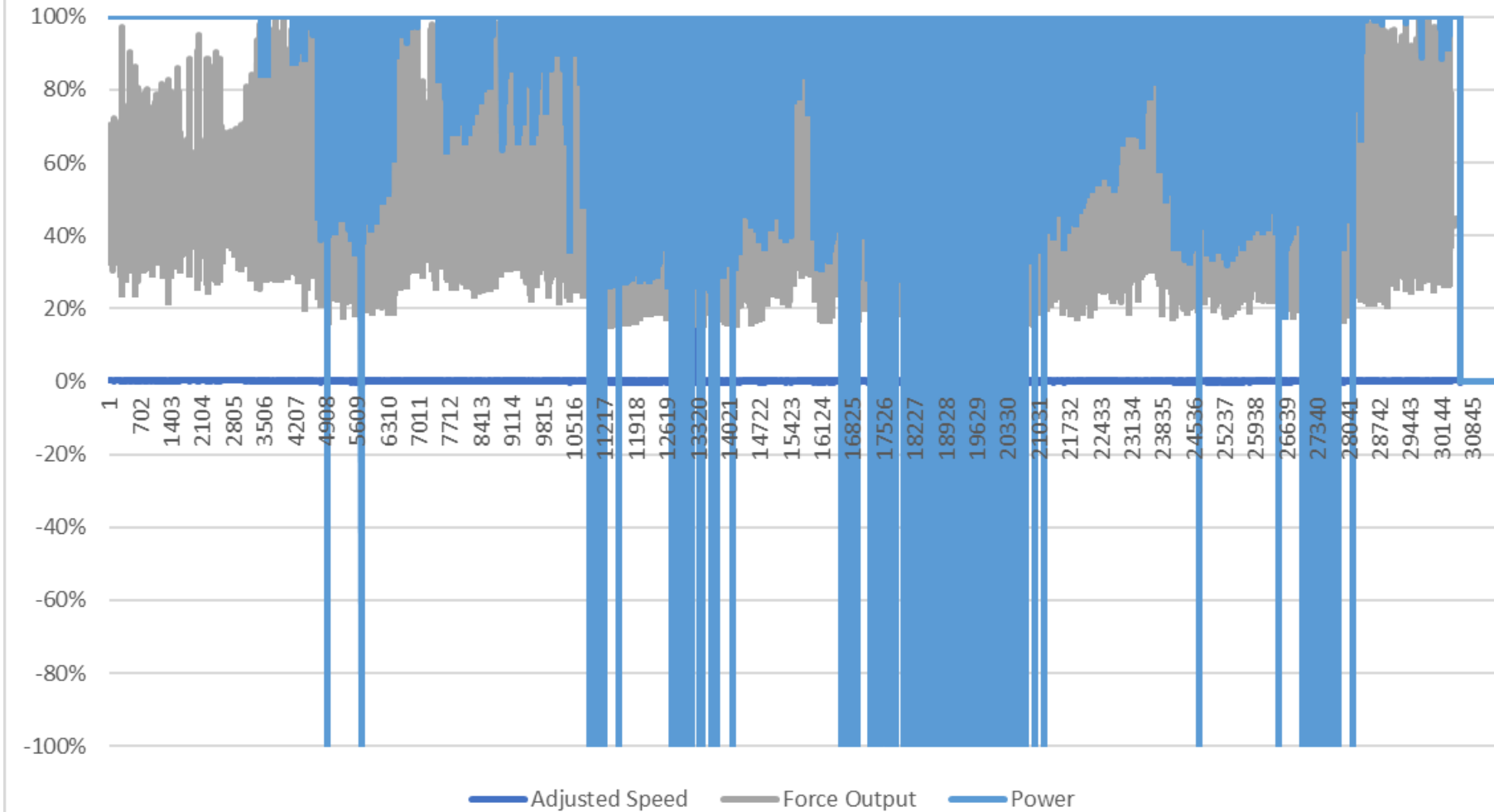
- Scoring based on same principal as RAG
 - Scores are provided for sections of the doctrine based on probability to be relevant to text
 - Scores have top 5 most relevant sections averaged
- Minor improvement in the relevance shown with use of RAG
- Responses from prompts about arbitrary topic include to show reference for responses with no relevance

Response	LLM with RAG	LLM without RAG	Random prompts
Score	0.941	0.921	0.873

- Using RAG, LLM performance has been shown to have minor increase in relevance to enemy doctrine
- Theoretical application of the DRON concept is acceptable for AI assisted OPFOR given DXTRs is modified to provide the requisite formatted data.



Performance Analysis



Snip

Segment	Point (fid)	Elevation	Slope%	Force Output	Power Output	Total Elevation Gain	Average Slope	Total Force Output
31	7	340	0	841.1764706	1051.470588	1950	5.6827	5859.1
2	6808	150	-6.666666667	870.1825558	1211.004057			
48	15528	310	12.90322581	790.1960784	771.0784314			
32	5417	720	51.38888889	669.2232694	105.7279818			
44	17970	170	-23.52941176	953.3333333	1668.333333			
37	7253	260	0	841.1764706	1051.470588			